

Shammas Ahmed

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EDUCATION

Rice University

B.S. Operations Research (specialization in Algorithms)

GPA: 3.92/4.0

Houston, TX

May 2027

Relevant Coursework

Linear and Integer Programming, Convex Optimization, Discrete Optimization, Large-Scale Optimization
Stochastic Optimization, Online Optimization, Machine Learning with Graphs, Algorithm Design, Simulation Modeling

EXPERIENCE

DSO Summer Scholar

Data Science & Operations Department, USC Marshall

May 2026 – August 2026

Los Angeles, CA

- Analyzed benchmarking and performance metrics across 24 papers on machine learning for optimization
- Established a constant additive “look-ahead premium” (LAP) in regret-based evaluation that distorts comparisons of model performance and benchmark difficulty
- Proposed a regret-reporting policy that accounts for LAP to reflect true model performance on given instances
- Research manuscript under review at the **NeurIPS MLxOR Workshop 2026**

Undergraduate Research Assistant - Mixed Integer Programming

Operations Research Department, Rice University

May 2025 – Dec 2025

Houston, TX

- Investigated a shortest-path problem with homology constraints (applications in robotics, healthcare, and transportation)
- Reformulated the problem as a network-flow optimization model and established NP-hardness
- Developed a polynomial-time heuristic and characterized instances for which the performance ratio is finite

Teaching Assistant - Introduction to Engineering Computation

Rice University School of Engineering and Computing, Rice University

August 2024 – Present

Houston, TX

- Deliver weekly 1-hour lectures on concepts in math and engineering (e.g. neural networks, first-order optimization methods, root-finding methods) with implementation in MATLAB and Python
- Lead weekly office hours, grade assignments, provide feedback for 20+ students

PROJECTS

Homologous Path Planner | Python, Gurobi

May 2025 – Dec 2025

- Designed optimization-based path planner that finds the shortest path under homology constraints
- Transformed paths into flows, implemented harmonic projection constraints on 20×20 grids
- Implemented the optimization model in Gurobi and visualized solutions in Python

Owl Help You Plan | Python, OR-Tools, Gurobi

February 2024 – April 2024

- Developed back-end optimization model for a vacation itinerary planning application
- Formulated as a Vehicle Routing Problem with constraints on destinations, travel time, cost and trip duration
- Implemented using OR-Tools and Gurobi, visualized on a local website using React

RESEARCH PRESENTATIONS

Regret Metrics in Decision-Focused Learning - DSO Summer Research Showcase

2026

The Inverse of the Integer Knapsack Problem - UT Arlington Math Symposium

2024

The Inverse of the Integer Knapsack Problem - Rice Summer Undergraduate Research Symposium

2024

SKILLS

Programming: Python, MATLAB, R

Optimization: Gurobi, JuMP, PyEPO, PuLP, OR-Tools

Developer Tools: Git, GitHub, VS Code, Google Colab

Methods: Mathematical modeling, algorithm design, data analysis, data visualization